

LCD ALARM CLOCK

UI & Component Design Specification

v3.0 · LVGL Reference

1. Overview

This document is the single source of truth for the LCD alarm clock UI. It covers hardware requirements, design tokens, a reusable component library, screen-by-screen composition, and the full state machine.

The device has no touch screen. All interaction is driven by physical buttons and a slide switch. The visual language is inspired by phosphor green VFD displays: high contrast, near-black background, minimal chrome.

1.1 Modularity principles

- Every visual element maps to exactly one named component.
- Components own their own padding, border, and typography — callers only set slot content.
- State variants (idle, editing, armed, disabled) are properties of components, not separate components.
- TopBar and BottomBar are permanent — mounted once, never hidden or recreated.

1.2 Screen structure

Every screen follows the same shell. Switching screens means changing ContentPanel slot contents only.

Slot	Content
TopBar	Always visible — date, wifi indicator, mode badge
ContentPanel	Variable — ContentPanelHeader / Body / Widget / Footer slots
BottomBar	Always visible — StatusStrip only (weather + now-playing)

1.3 Pseudo-structure

```
<TopBar />
<ContentPanel>
  <ContentPanelHeader>{label | nil}</ContentPanelHeader>
  <ContentPanelBody>
    <ClockDisplay blink={on|off} />
  </ContentPanelBody>
  <ContentPanelWidget>
    <DayRow activeDay={...} />  {/* nil on most screens */}
  </ContentPanelWidget>
  <ContentPanelFooter>{hint | alarm | dismiss}</ContentPanelFooter>
</ContentPanel>
<BottomBar />
```

2. Hardware & Screen

2.1 Recommended screen

3.5"–4.0" IPS LCD, 480×320 px landscape.

2.2 Button map

Button	Location	Function
MON–SUN (×7)	Front	Select/switch active day in auto alarm edit. Press current selected day to toggle armed/disarmed.
TIME	Top	Enter clock time edit mode. Press again to confirm.
HR+	Top	Increment hour. Active during EDIT_CLOCK, SET_SINGLE, SET_AUTO_EDIT.
MIN+	Top	Increment minute. Same context sensitivity as HR+.
SET	Top	Start alarm set flow. Press again to confirm. Behaviour depends on mode.
DISMISS	Top (large)	Dismiss firing alarm immediately.
MODE switch	Side	SINGLE AUTO — changes active alarm mode.
PREV / PLAY / SKIP	Side	Spotify media controls.

3. Design Tokens

3.1 Colors

	#0A0C0F	Background	Screen background — all screens
	#0D1117	Surface	Card / tile backgrounds
	#1AFF8C	Primary Green	Active digits, selected states, armed indicator
	#0E8A4A	Mid Green	AM/PM label
	#0E6A3A	Dim Green	Colon, status text, context labels, hint keywords
	#1A3A28	Border Active	Armed tile border, selected state
	#1A2520	Border Default	Default tile / card borders

	#2A5040	Muted Text	Date label, captions
	#1A7A4A	Status Text	Weather and track labels
	#1A4030	Hint Text	Footer hint text
	#F09595	Alert Red	Firing alert header
	#EF9F27	Amber	Dismiss footer text
	#854F0B	Amber Dim	Disabled hint text in header

3.2 Typography

Token	Value
--orbitron-xl	Orbitron Bold 88px — ClockDisplay idle
--orbitron-lg	Orbitron Bold 82px — ClockDisplay edit / firing
--orbitron-md	Orbitron Bold 20px — ContentPanelHeader alert style
--orbitron-xs	Orbitron Regular 9px — DayTile labels
--orbitron-ampm	Orbitron Bold 17px — AM/PM superscript
--mono-md	Share Tech Mono 11px — StatusStrip, DateLabel
--mono-sm	Share Tech Mono 10px — ContentPanelFooter hints
--mono-xs	Share Tech Mono 9px — ModeBadge, DayTile times, CPHeader hint suffix

3.3 Spacing

Token	Value
--screen-pad-h	22px left/right
--screen-pad-v-top	16px top
--screen-pad-v-bot	14px bottom
--strip-pad	5–6px vertical (status strip)
--tile-gap	5px between DayTiles
--tile-pad	6–8px top/bottom · 6–8px left/right (DayTile)

3.4 Style Application Rules

Apply these properties precisely. No additional effects (shadow, gradient, glow, outline) unless visible in reference images.

Element / State	Exact Style Values
ClockDisplay – idle	Orbitron Bold 88px · #1AFF8C · letter-spacing -3px · centered · colon #0E6A3A
ClockDisplay – edit/firing	Orbitron Bold 82px · #1AFF8C · entire lv_obj opacity blinks 1.0 ↔ 0.25 at 500ms
ClockDisplay – disarmed edit	Same as edit · base opacity reduced to ~0.3 (disarmed day selected)
AM/PM superscript	Orbitron Bold 17px · #0E8A4A · vertical-align super · margin-left 5px
CPHeader – normal	Share Tech Mono 11px · #0E6A3A · tracking 0.18em · centered · padding 4px top 2px bot
CPHeader – alert (FIRING)	Orbitron Bold 20px · #F09595 · tracking 0.08em · centered
CPHeader hint suffix	Share Tech Mono 9px · #1A4030 (armed) or #854F0B (disarmed) · appended after day label
CPFooter – hint	Share Tech Mono 10px · #1E4030 · centered · keyword spans #0E6A3A
CPFooter – alarm	Share Tech Mono 11px · label #1A7A4A · time #1AFF8C · armed dot 6×6px #1AFF8C
CPFooter – dismiss	Share Tech Mono 11px · #EF9F27 · tracking 0.12em · centered · no border
DayTile – default	bg #0D1117 · border 0.5px #1A2520 · label #2A4535 · time #1E3A2A · pip #1A2A20
DayTile – has-alarm	bg #0D1510 · border 0.5px #1A3A28 · label/time/pip #0E6A3A
DayTile – selected armed	bg #0A1F10 · border 0.5px #1AFF8C · label/time #1AFF8C · pip #1AFF8C · clock blinks
DayTile – selected disarmed	Same as selected armed + lv_obj opacity 45% · pip #1A2A20
StatusStrip borders	0.5px solid #111A14 top and bottom
StatusStrip divider	0.5px solid #1A2A20 · height 13px · vertically centered
ModeBadge – SINGLE	text #888780 · bg #1A1A18 · border 0.5px #2A2A28 · border-radius 4px
ModeBadge – AUTO	text #1AFF8C · bg #0D1A12 · border 0.5px #1A4028 · border-radius 4px

3.5 Text Formatting Rules

Exact string formats for all dynamic UI text. Implement these precisely to avoid implementation-defined variations.

Element	Format
DateLabel	'TUESDAY · MARCH 17 · 2026' — uppercase weekday · middot separators (·) · 4-digit year

CPFooter alarm — SINGLE	'alarm 6:30 AM armed' — no leading zero on hour
CPFooter alarm — AUTO	'next alarm WED 7:15 AM' — 3-char uppercase day · 12-hour no leading zero
CPFooter — no alarm set	nil text — footer hidden entirely (LV_OBJ_FLAG_HIDDEN) when no alarm is configured and none are armed
CPHeader auto edit — armed	'— MON ● —' — em-dashes · spaces · filled circle U+25CF
CPHeader auto edit — disarmed	'— MON ○ —' — em-dashes · spaces · empty circle U+25CB
CPHeader hint — armed	'press again to disable' — Share Tech Mono 9px #1A4030
CPHeader hint — disarmed	'disabled · press again to enable' — #854F0B
StatusStrip weather	'✱ 72°F Clear' — Unicode sun U+2600 · degree symbol · condition word
StatusStrip track	'♪ Track Name · Artist II' — U+266B · separator · U+2016 pause
StatusStrip truncation	Truncate with ... if track or artist exceeds ~140px at 11px font
DayTile time — set	'6:30' — no leading zero on hour · colon separator
DayTile time — unset	'——' — two em-dashes U+2014

4. Component Library

9 components cover the entire UI. Each is an lv_obj subtree created once and reconfigured across screens.

C01 · TopBar

Permanent header. Never hidden.

C01	TopBar
Persistent header — date · wifi · mode badge	
Property	Value
Height	28px
Children	C07 DateLabel (flex:1) + C09 WiFiIndicator + C08 ModeBadge
LVGL	lv_obj, LV_LAYOUT_FLEX, LV_FLEX_FLOW_ROW, fixed top

Used in: ALL screens

C02 · ContentPanel + sub-components

Variable main area. Four named sub-components as slots. The panel itself is permanent — only slot content changes.

C02	ContentPanel
Main content area — sub-component slots swapped per screen	
Property	Value
Layout	flex column, flex:1
Sub-components	C02a–d below
LVGL	lv_obj, LV_LAYOUT_FLEX, LV_FLEX_FLOW_COLUMN

Used in: ALL screens

C02a · ContentPanelHeader

Context label or alert heading. Hidden on idle screens.

Property	Value
Normal style	Share Tech Mono 11px, #0E6A3A, tracking 0.18em, centered
Alert style	Orbitron Bold 20px, #F09595, tracking 0.08em, centered
Hint suffix	Share Tech Mono 9px — appended after label; #1A4030 (normal) or #854F0B (disabled)
Armed indicator	'●' (filled, #1AFF8C) or '○' (empty) appended to day name when in auto edit
Hidden	LV_OBJ_FLAG_HIDDEN when nil

Used in: EDIT_CLOCK, SET_SINGLE, SET_AUTO_EDIT, FIRING

C02b · ContentPanelBody

Always-visible center slot. Holds ClockDisplay and expands to fill remaining vertical space.

Property	Value
Layout	flex:1, centers child
Content	Always C03 ClockDisplay

Used in: ALL screens

C02c · ContentPanelWidget

Optional slot below the body. Holds DayRow in auto alarm edit. Hidden on all other screens.

Property	Value
Content	C05 DayRow when active, nil otherwise
Hidden	LV_OBJ_FLAG_HIDDEN when nil

Used in: SET_AUTO_EDIT only

C02d · ContentPanelFooter

Always present. Three variants depending on screen context.

Property	Value
Variant — hint	Share Tech Mono 10px, #1E4030, centered. Keywords in #0E6A3A
Variant — alarm	Alarm icon + armed dot + label + time in #1AFF8C
Variant — dismiss	Share Tech Mono 11px, #EF9F27 — plain text 'DISMISS to stop'

Used in: ALL screens

C03 · ClockDisplay

Time readout. Reused across every screen — idle shows live RTC time, edit flows show the value being edited, firing shows the alarm time. The entire display blinks as a unit during any edit state.

C03	ClockDisplay HH:MM AM/PM — present on every screen
Property	Value
Idle font	Orbitron Bold 88px, #1AFF8C, tracking -3px
Edit/firing font	Orbitron Bold 82px (slightly smaller to make room for header)
AM/PM	Orbitron Bold 17px, #0E8A4A, vertical-align super
Colon	#0E6A3A
blink param	on off — whole display toggles opacity 1.0 ↔ 0.25 at 500ms
LVGL	lv_label; lv_timer drives blink toggle on lv_obj opacity

Used in: ALL screens (via C02b)

C04 · BottomBar

Permanent footer. Always contains StatusStrip only. Alarm status has moved to ContentPanelFooter, so BottomBar is unconditionally simple.

C04	BottomBar
Persistent footer — StatusStrip only	
Property	Value
Children	C06 StatusStrip only — never toggled
LVGL	lv_obj fixed at bottom

Used in: ALL screens

C05 · DayRow + DayTile

Horizontal row of 7 DayTile sub-components. Used in ContentPanelWidget during auto alarm editing.

C05	DayRow
7 × DayTile — auto alarm day navigation and arm toggle	
Property	Value
Layout	flex row, gap 5px, justify-content: center
activeDay param	MON–SUN — which tile is selected; defaults to MON on entry
LVGL	lv_obj, LV_LAYOUT_FLEX, LV_FLEX_FLOW_ROW

Used in: SET_AUTO_EDIT (via C02c)

C05a · DayTile

Individual tile. Four visual states:

State	Appearance
default	bg #0D1117, border #1A2520, label/time #2A4535/#1E3A2A, pip #1A2A20
has-alarm	bg #0D1510, border #1A3A28, label/time/pip #0E6A3A
selected (armed)	bg #0A1F10, border #1AFF8C, label/time/pip #1AFF8C, clock blinks
selected (disarmed)	selected styles at 45% opacity, pip #1A2A20 — alarm disabled for this day

Used in: SET_AUTO_EDIT (via C05)

C06 · StatusStrip

Weather + now-playing. Always visible via BottomBar. Contains two items divided by a vertical rule.

C06	StatusStrip
Weather + now-playing — always in BottomBar	
Property	Value
Height	~26px, 0.5px borders top/bottom

Weather	Sun icon + temp #1AFF8C + condition #1A7A4A, Share Tech Mono 11px
Now playing	Music note + track #1AFF8C + artist #1A4030 + pause icon II
Divider	0.5px solid #1A2A20, 13px tall

Used in: ALL screens (via C04)

C07 • DateLabel

Centered date string in TopBar. Share Tech Mono 11px, #2A5040, tracking 0.16em. Format: 'TUESDAY · MARCH 17 · 2026'.

Used in: ALL screens (via C01)

C08 • ModeBadge

Pill label in TopBar reflecting the MODE switch. SINGLE: text #888780, bg #1A1A18. AUTO: text #1AFF8C, bg #0D1A12. Updates on GPIO interrupt.

Used in: ALL screens (via C01)

C09 • WiFiIndicator

Three-arc SVG in TopBar. Inner two arcs #1AFF8C, outer arc #0E6A3A (dim). 18×14px.

Used in: ALL screens (via C01)

5. UI States

Six screens compose the full UI. Each entry shows the rendered screen followed by its component composition table.

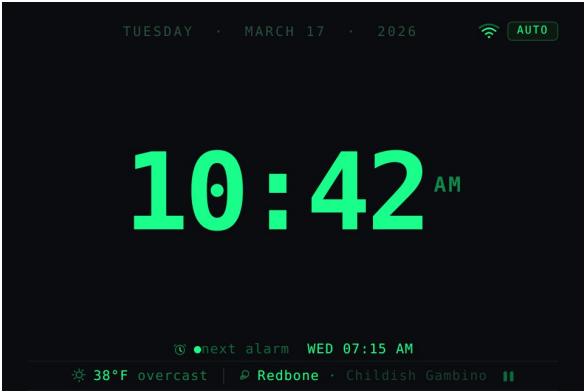
5.1 Idle — Single & Auto

The default screen. ContentPanel has no header — clock fills the body, next alarm in the footer.

Idle — Single mode



Idle — Auto mode



Component	ID	Slot / State
TopBar	C01	Date · WiFi · ModeBadge (SINGLE grey / AUTO green)
CPHeader	C02a	nil — hidden
CPBody	C02b	C03 ClockDisplay, blink: off — live RTC time
CPWidget	C02c	nil
CPFooter	C02d	Alarm variant — icon + dot + 'alarm HH:MM' or 'next alarm DAY HH:MM'
BottomBar	C04	C06 StatusStrip

5.2 Edit Clock Time

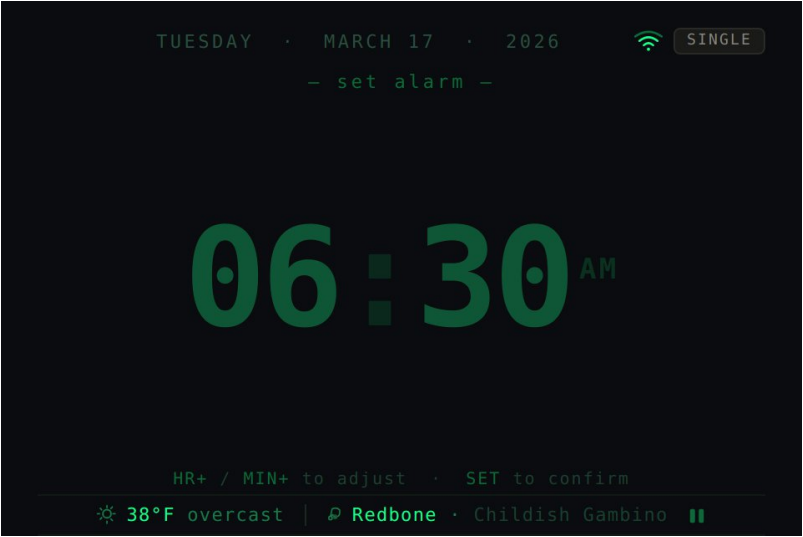
Entered by pressing TIME. Press TIME again to confirm. HR+ and MIN+ adjust the time. Entire clock blinks.



Component	ID	Slot / State
CPHeader	C02a	'— set time —' (normal style)
CPBody	C02b	C03 ClockDisplay, blink: on
CPWidget	C02c	nil
CPFooter	C02d	Hint: 'HR+ / MIN+ to adjust · TIME to confirm'

5.3 Set Alarm — Single Mode

Entered by pressing SET in SINGLE mode. Press SET again to confirm.



Component	ID	Slot / State
CPHeader	C02a	'— set alarm —' (normal style)
CPBody	C02b	C03 ClockDisplay, blink: on
CPWidget	C02c	nil
CPFooter	C02d	Hint: 'HR+ / MIN+ to adjust · SET to confirm'

5.4 Set Alarm — Auto Mode

Entered by pressing SET in AUTO mode. Defaults to Monday. Press a different day button to switch. Press the currently selected day button to toggle it armed/disarmed.

Armed state (● = alarm on)



Disarmed state (○ = alarm off)



Component	ID	Slot / State
CPHeader	C02a	'— {Day} ● —' armed, or '— {Day} ○ —' disarmed + hint suffix
CPBody	C02b	C03 ClockDisplay, blink: on (very dim when day is disarmed)
CPWidget	C02c	C05 DayRow — activeDay = MON on entry; selected tile reflects arm state
CPFooter	C02d	Hint: 'HR+ / MIN+ to adjust · SET to confirm'

Pressing the currently selected day button toggles its armed state. The header indicator and tile pip update immediately. HR+/MIN+ still adjust the time even when disabled — the time is preserved so re-enabling it restores the previous value.

5.5 Alarm Firing

When a firing condition is met. Clock shows the alarm time. No day label. Rings until DISMISS.



Component	ID	Slot / State
CPHeader	C02a	'!! ALARM !!' (alert style — Orbitron, #F09595)
CPBody	C02b	C03 ClockDisplay, blink: off — shows firing alarm time
CPWidget	C02c	nil
CPFooter	C02d	Dismiss variant: 'DISMISS to stop' in #EF9F27

6. State Machine

Every state, trigger, resulting state, and action. The MODE switch is ambient context — not a state transition.

State	Trigger	Next State	Action / Notes
IDLE	TIME	EDIT_CLOCK	Enter clock edit; begin blinking full display
IDLE	SET (SINGLE mode)	SET_SINGLE	Show alarm editor; begin blinking
IDLE	SET (AUTO mode)	SET_AUTO_EDIT	Open on Monday by default; show DayRow; begin blinking
IDLE	MODE switch flipped	IDLE	Update ModeBadge; re-evaluate next alarm
IDLE	Alarm time reached + armed	FIRING	Play alarm tone; show alert screen
EDIT_CLOCK	HR+	EDIT_CLOCK	Increment hour; wrap 12 → 1
EDIT_CLOCK	MIN+	EDIT_CLOCK	Increment minute; wrap 59 → 0
EDIT_CLOCK	TIME	IDLE	Commit new clock time; stop blink
SET_SINGLE	HR+	SET_SINGLE	Increment alarm hour; wrap 12 → 1
SET_SINGLE	MIN+	SET_SINGLE	Increment alarm minute; wrap 59 → 0
SET_SINGLE	SET	IDLE	Save alarm time; return to idle
SET_AUTO_EDIT	Day button (other)	SET_AUTO_EDIT	Switch active day; update header + DayRow
SET_AUTO_EDIT	Day button (current)	SET_AUTO_EDIT	Toggle arm/disarm for active day; update header indicator + pip
SET_AUTO_EDIT	HR+	SET_AUTO_EDIT	Increment alarm hour for active day
SET_AUTO_EDIT	MIN+	SET_AUTO_EDIT	Increment alarm minute for active day
SET_AUTO_EDIT	SET	IDLE	Save active day; return to idle
FIRING	DISMISS	IDLE	Stop alarm tone; clear alert screen

6.1 Mode switch behaviour

- SINGLE: TIME edits clock. SET edits the one global alarm. Day buttons inert.
- AUTO: TIME edits clock. SET opens alarm editor on Monday. Day buttons navigate and toggle arm state.
- If MODE switch is flipped mid-edit, the flow cancels and returns to IDLE.

6.2 Alarm armed logic

- SINGLE: armed when a time is set and alarm switch is ON.
- AUTO: each day has an independent armed flag. Toggled by pressing the selected day button. A disarmed day retains its time — re-arming restores it.
- The next alarm footer always shows the nearest upcoming armed slot across all active days.
- There is no auto-dismiss. The alarm rings until DISMISS is pressed.

6.3 Long-press TIME — AM/PM toggle

In all edit states (EDIT_CLOCK, SET_SINGLE, SET_AUTO_EDIT): a long-press of the TIME button (≥ 800 ms) toggles AM/PM. The 17px superscript label updates live. Short-press TIME still confirms and exits the edit state as normal.

7. Component Reuse Matrix

Green dot = component present on that screen. C02b and C03 are present on every screen — the clock never disappears.

ID	Component	IDLE	EDIT CLOCK	SET SINGLE	SET_AUTO EDIT	FIRING
C01	TopBar					
C02	ContentPanel					
C02a	↳ CPHeader					
C02b	↳ CPBody					
C02c	↳ CPWidget					
C02d	↳ CPFooter					
C03	ClockDisplay					
C04	BottomBar					
C05	DayRow					
C05a	↳ DayTile ×7					
C06	StatusStrip					
C07	DateLabel					
C08	ModeBadge					
C09	WiFiIndicator					

Appendix · Component Renders & Dimensions

Each component rendered at 2× resolution showing all visual states. Dimension tables specify exact pixel values for 480×320 px target. No additional effects (shadow, gradient, glow) beyond what is shown.

C01 · TopBar

TUESDAY · MARCH 17 · 2026



AUTO

Property	Value
Width × Height	480 × 28 px
Background	#0A0C0F
Padding	0 horizontal sides: 22px · vertical: centered
Layout	flex row · space-between
Left spacer	28px fixed (balances wifi+badge group on right)
C07 DateLabel	flex:1 · centered · Share Tech Mono 11px · #2A5040 · tracking 0.16em · uppercase
C09 WiFiIndicator	18×14px · inner arcs #1AFF8C · outer arc #0E6A3A
C08 ModeBadge	Share Tech Mono 9px · border-radius 4px · see below

SINGLE

AUTO

ModeBadge state	Values
SINGLE	text #888780 · bg #1A1A18 · border 0.5px #2A2A28
AUTO	text #1AFF8C · bg #0D1A12 · border 0.5px #1A4028
Padding	2px vertical · 7px horizontal

C03 · ClockDisplay — idle and edit

10 : 42^{AM}

↑ Idle (blink off)

10:42 AM

↑ Edit / Firing (blink on — shown at 25% opacity phase)

State	Font · Size · Color · Notes
Idle	Orbitron Bold · 88px · #1AFF8C · tracking -3px · colon #0E6A3A
Edit / Firing	Orbitron Bold · 82px · #1AFF8C · lv_obj opacity toggles 1.0 ↔ 0.25 at 500ms
AM/PM (both)	Orbitron Bold · 17px · #0E8A4A · vertical-align super · margin-left 5px
Long-press TIME	≥800ms in any edit state → toggles AM/PM live

C02a · ContentPanelHeader — all variants

Hidden (LV_OBJ_FLAG_HIDDEN) on idle screens. Three style variants.

— set time —
— Monday ● — press again to disable
!! ALARM !!

Variant	Font · Color · Notes
Normal (edit)	Share Tech Mono 11px · #0E6A3A · tracking 0.18em · centered · padding 4px top 2px bot
Auto edit armed	Normal + ' ● ' (U+25CF, #1AFF8C) after day name · hint suffix #1A4030
Auto edit disarmed	Normal + ' ○ ' (U+25CB, #1AFF8C) after day name · hint suffix #854F0B
Alert (FIRING)	Orbitron Bold 20px · #F09595 · tracking 0.08em · centered

C02d · ContentPanelFooter — all variants

Never hidden. Content and color change per screen context.

HR+ / MIN+ to adjust · SET to confirm

🕒 ● next alarm WED 7:15 AM

DISMISS to stop

Variant	Spec
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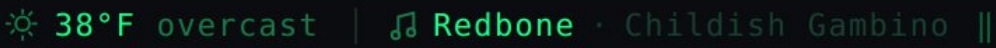
Hint (edit screens)	Share Tech Mono 10px · #1E4030 · centered · HR+/MIN+/SET/TIME keywords in #0E6A3A
Alarm — SINGLE	'alarm H:MM AM armed' · STM 11px · label #1A7A4A · time #1AFF8C · armed dot 6×6px #1AFF8C
Alarm — AUTO	'next alarm DAY H:MM AM' · same styling · 3-char uppercase day
No alarm set	LV_OBJ_FLAG_HIDDEN — footer not shown when no alarm is configured
Dismiss (FIRING)	'DISMISS to stop' · STM 11px · #EF9F27 · tracking 0.12em · centered · no border
Padding	4px top · 3px bottom

C05 · DayRow + C05a · DayTile — all states

Left-to-right in the render: default · has-alarm · selected-armed · selected-disarmed · has-alarm · default · default.

	
DayRow	flex row · gap 4px · justify-content: center · padding 6px top 4px bottom
DayTile size	min-width 46px · border-radius 6px · padding 6px top/bottom · 6px left/right
Day label	Orbitron Regular 9px · uppercase
Time	Share Tech Mono 10px · 'H:MM' or '——' (U+2014 ×2) if unset
Pip	4×4px circle · bottom-center
default	bg #0D1117 · border 0.5px #1A2520 · label #2A4535 · time #1E3A2A · pip #1A2A20
has-alarm	bg #0D1510 · border 0.5px #1A3A28 · label/time/pip #0E6A3A
selected — armed	bg #0A1F10 · border 0.5px #1AFF8C · label/time/pip #1AFF8C · clock blinks
selected — disarmed	Same as armed · lv_obj opacity 45% · pip #1A2A20

C06 · StatusStrip

	
Property	Value
Width × Height	480 × 26px
Borders	0.5px solid #111A14 top and bottom
Divider	0.5px solid #1A2A20 · height 13px · vertically centered
Padding per item	0 12px horizontal

Weather	☀ icon 13×13px · temp STM 11px #1AFF8C · condition #1A7A4A
Now playing	🎵 icon 11×11px · track STM 11px #1AFF8C · '.' · artist #1A4030 · pause 12×12px
Truncation	... (U+2026) if track or artist exceeds ~140px at 11px